REGULARITY OF A GENERATING SEQUENCE DOES NOT DEPEND ON THE CHOSEN GENERATORS OF THE IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Quasi-regularity, H_1-regularity and Koszul-regularity of a sequence, and the equivalences between them for sequences generating the ideal. the author cites Lemma lemma-H1-regular-quasi-regular at proof lines 7740-7742. Supplies the reduction of the last two parts to the quasi-regular case.
- The change-of-basis matrix between two generating sequences induces a map of Koszul complexes which is an isomorphism after inverting its determinant. the author cites Lemma lemma-functorial at proof lines 7730-7733. Produces the comparison map between the two Koszul complexes.
- The cohomology modules of the Koszul complex of a sequence generating the ideal are annihilated by the ideal. the author cites Lemma lemma-homotopy-koszul at proof lines 7734-7736. Upgrades the local isomorphism to a quasi-isomorphism over the whole ring.

Source reference. Original lemma 066A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7929–7929. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0624: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7783–7783. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0663: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7922–7922. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 062I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7797–7797. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let I be an ideal generated by $f_1, \ldots, f_r \in R$.*

1. *If I can be generated by a quasi-regular sequence of length r, then $f_1, \ldots, f_r$ is a quasi-regular sequence.*
2. *If I can be generated by an H_1-regular sequence of length r, then $f_1, \ldots, f_r$ is an H_1-regular sequence.*
3. *If I can be generated by a Koszul-regular sequence of length r, then $f_1, \ldots, f_r$ is a Koszul-regular sequence.*

Proof. If I can be generated by a quasi-regular sequence of length r, then I/I^2 is free of rank r over R/I. Since $f_1, \ldots, f_r$ generate by assumption we see that the images $\overline{f}_i$ form a basis of I/I^2 over R/I. It follows that $f_1, \ldots, f_r$ is a quasi-regular sequence as all this means, besides the freeness of I/I^2, is that the maps $\mathrm{Sym}^n_{R/I}(I/I^2) \to I^n/I^{n+1}$ are isomorphisms.

We continue to assume that I can be generated by a quasi-regular sequence, say $g_1, \ldots, g_r$. Write $g_j = \sum a_{ij} f_i$. As $f_1, \ldots, f_r$ is quasi-regular according to the previous paragraph, we see that $\det(a_{ij})$ is invertible mod I. The matrix a_{ij} gives a map $R^{\oplus r} \to R^{\oplus r}$ which induces a map of Koszul complexes $\alpha : K_\bullet(R, f_1, \ldots, f_r) \to K_\bullet(R, g_1, \ldots, g_r)$, see Lemma 0624. This map becomes an isomorphism on inverting $\det(a_{ij})$. Since the cohomology modules of both $K_\bullet(R, f_1, \ldots, f_r)$ and $K_\bullet(R, g_1, \ldots, g_r)$ are annihilated by I, see Lemma 0663, we see that α is a quasi-isomorphism.

Now assume that $g_1, \ldots, g_r$ is a H_1-regular sequence generating I. Then $g_1, \ldots, g_r$ is a quasi-regular sequence by Lemma 062I. By the previous paragraph we conclude that $f_1, \ldots, f_r$ is a H_1-regular sequence. Similarly for Koszul-regular sequences. $\square$